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API-driven Cloud Native Solutions

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CC ZG506

Lecture No. 14

Agenda

IoT

- IoT Basics
- Edge Computing
- IoT Platforms Overview
- OpenRemote IoT Platform
- SmartCity Implementation use case demo

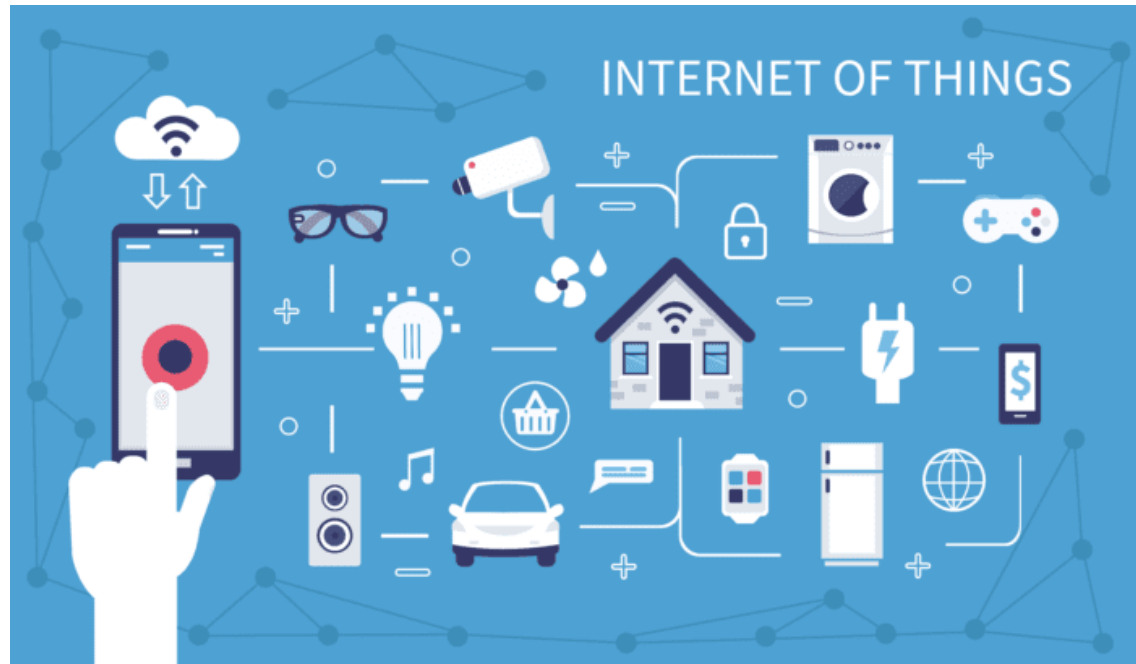


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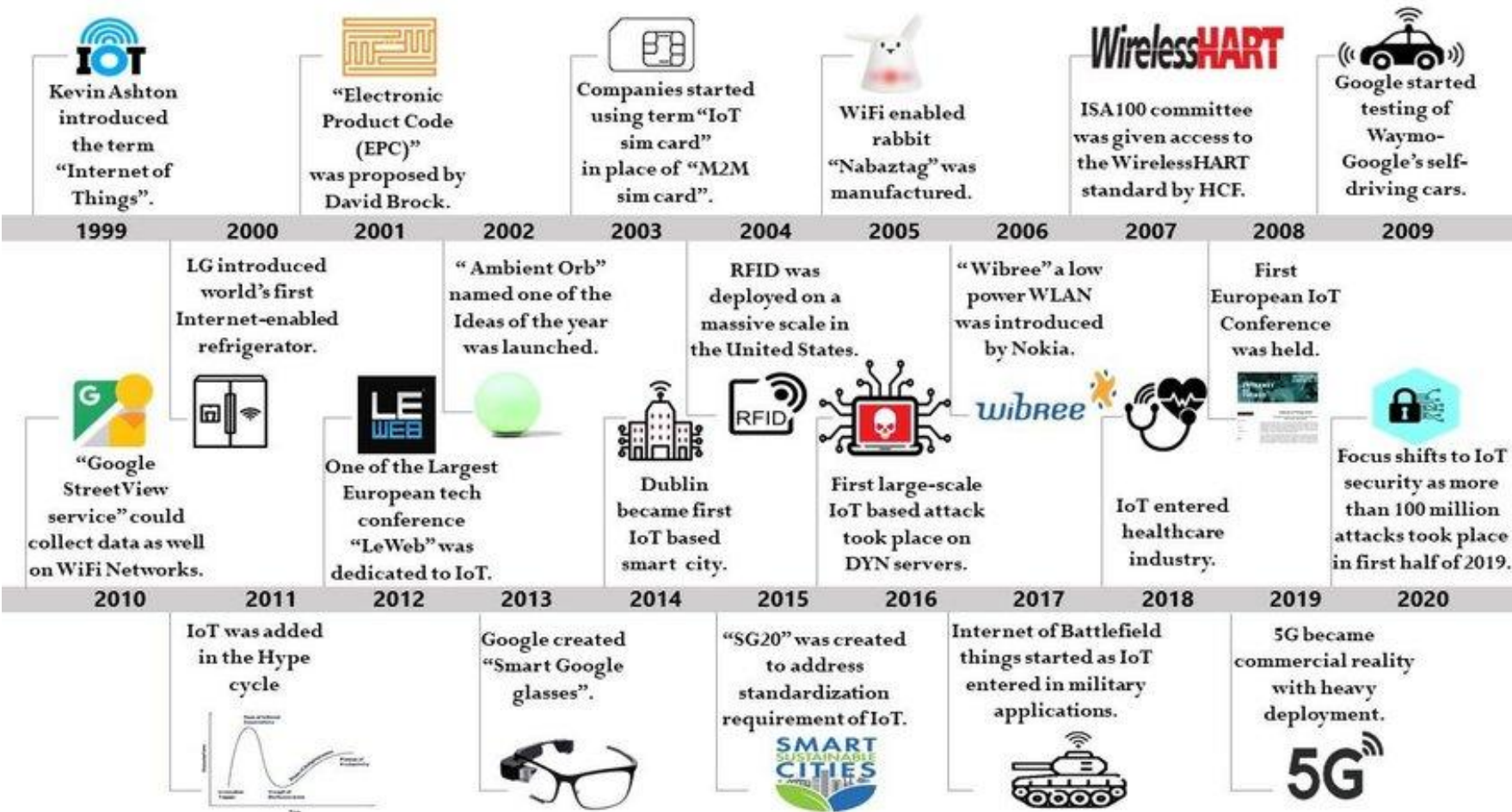


Internet of Things (IoT)

- Refers to the network of connected devices, sensors, and other physical objects that gather and share data.
- Examples include smart home devices, industrial sensors, wearable health trackers, and smart city infrastructure.

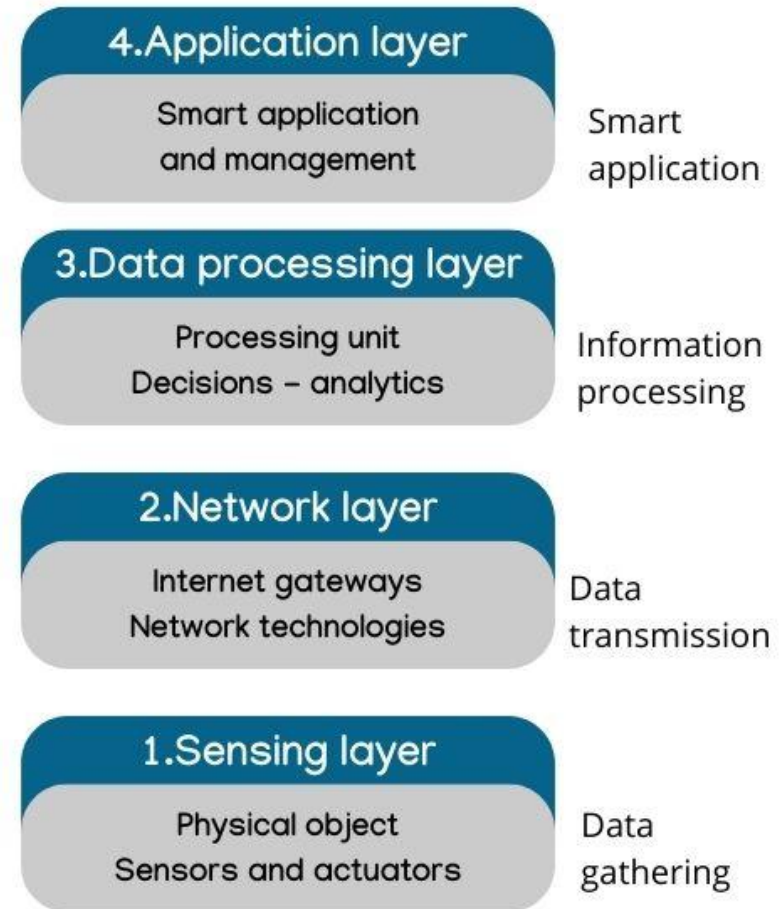


IoT Evolution



IoT Architecture

IoT architecture requires connected hardware (that gathers data), a means of communication to transmit the data to or from a cloud, a system that allows to process all data, and a data center (which can work in the cloud or in a local server) that will allow to store, handle, and analyze the information provided.



<https://www.deepseadev.com/en/blog/iot-architecture-explained/>

1. Sensing Layer

- This layer of IoT architecture contains all the different devices and sensors that are in charge of collecting data (from surroundings or specific tasks).

Examples:

- Environment sensors: Temperature, Humidity, Light, Air Quality
- Motion and Proximity sensors: Infrared (IR), Ultrasonic
- Wearable sensors: Heart rate monitor, accelerometers, electro dermal activity sensors
- Industrial sensors: Pressure, Gas, Vibration
- Smart Home Devices: Smart Thermostats, Smart Door Locks, Voice Assistants (Amazon Echo)

1. Sensing Layer

Examples:

- Healthcare devices: Glucose monitors, ECG monitors
- Agricultural sensors: Soil moisture sensors, pH sensors
- Vehicle sensors: Tire pressure, Fuel level, GPS sensors

These devices in the sensing layer gather data that flows to other layers in the IoT architecture for further processing, analysis, and action.

2. Network Layer

- This layer is responsible for the data transfer between the devices and the internet, this is made usually through a gateway that is in charge of adding additional processes.
- In this layer, the most important part is protocol configuration, data encryption, malware protection, and authentication.

3. Data Processing Layer

- This IoT architecture layer is in charge of managing all data analysis.
- This layer is very useful for IoT analytics. Another important aspect to consider, the data processing layer can be located in the gateway or in the cloud.
- Perform data cleaning, transformation, aggregation, and analysis here
- Can be on the Edge (Ex: IoT Gateway)

4. Application Layer

- This is where the data is processed and **shown to the end-user**. This IoT layer is usually located in the cloud.
- The application displays the data, and decisions can be made according to the information provided.

Example 1: Wearable Fitness Tracker

1. **Sensing Layer:** The wearable device has sensors like accelerometers, heart rate monitors, and GPS, which gather data on physical activity, heart rate, and location.
2. **Network Layer:** The data collected by the sensors is transmitted to a smartphone or cloud server via Bluetooth or Wi-Fi.
3. **Data Processing Layer:** Activities like data cleaning (e.g., removing noise, filling in missing GPS data), transformation (e.g., converting step counts into distance or speed), and aggregation (e.g., summarizing hourly heart rate trends). Sends data to the cloud / Edge.
4. **Application Layer:** The wearable app (on the smartphone) provides users with detailed insights, exercise goals, and health tracking, displaying data in an easy-to-understand format. It may also offer alerts or reminders for activity based on the processed data.

Example 2: Air Pollution Monitoring System

1. **Sensing Layer:** Air quality sensors are deployed in various locations to measure pollutants like CO₂, particulate matter (PM_{2.5}), and other gases in the air.
2. **Network Layer:** The sensors transmit collected data to a central server through cellular, or Wi-Fi networks.
3. **Data Processing Layer:** Data from multiple sensors is aggregated, analyzed, and processed to determine pollution levels in real-time. Analytics can also forecast pollution trends.
4. **Application Layer:** The processed data is used to provide information to users through mobile apps or websites, displaying air quality levels, issuing alerts, and offering recommendations on outdoor activities based on pollution levels.

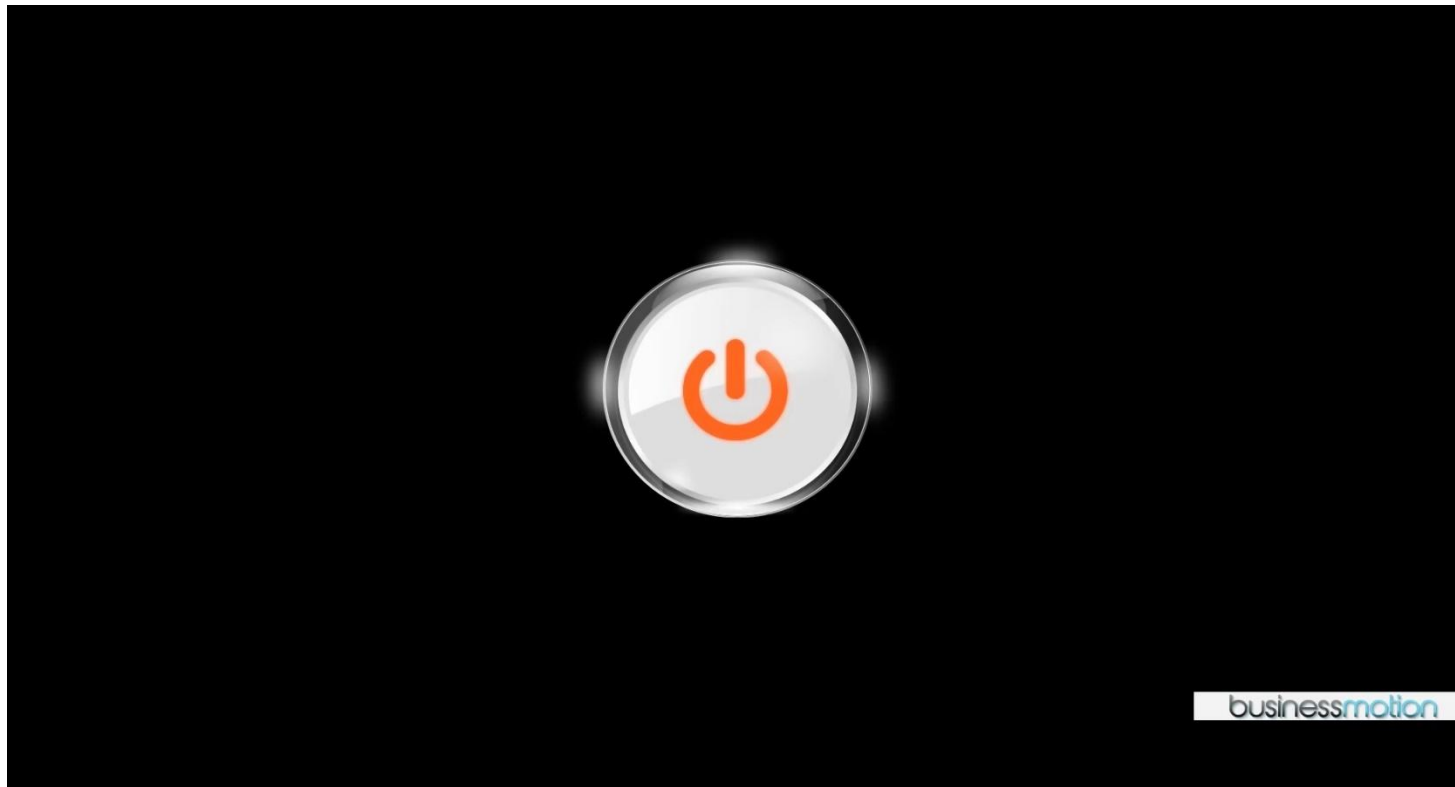


Edge Computing [Only Basics]

Edge Computing

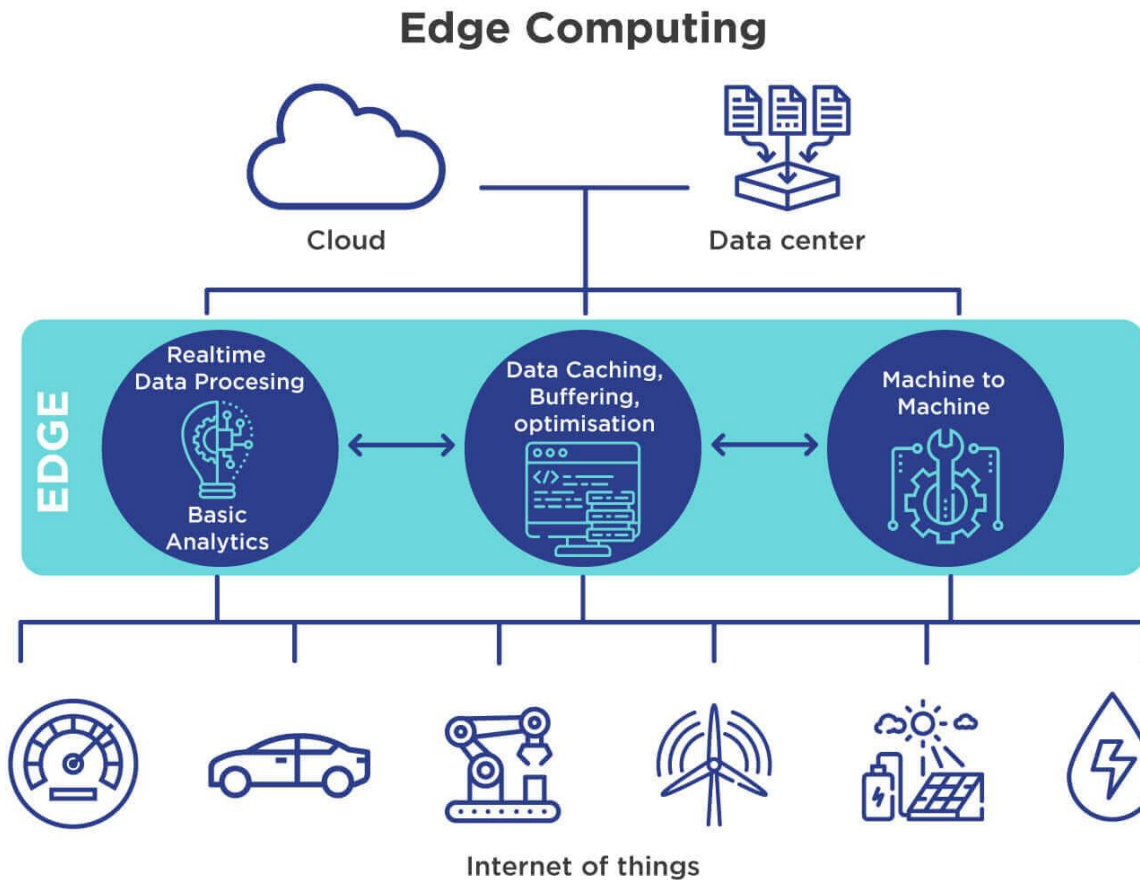
- Edge computing is a distributed information technology (IT) architecture in which client data is processed at the periphery of the network, as close to the originating source as possible.
- Edge computing is the **deployment of computing and storage resources** at the **location where data is produced**.
- It helps to provide server resources, data analysis, and artificial intelligence to data collection sources and cyber physical sources like smart sensors and actuators.

Edge Computing



<https://www.youtube.com/watch?v=cEOUeItHDdo>

Edge Computing



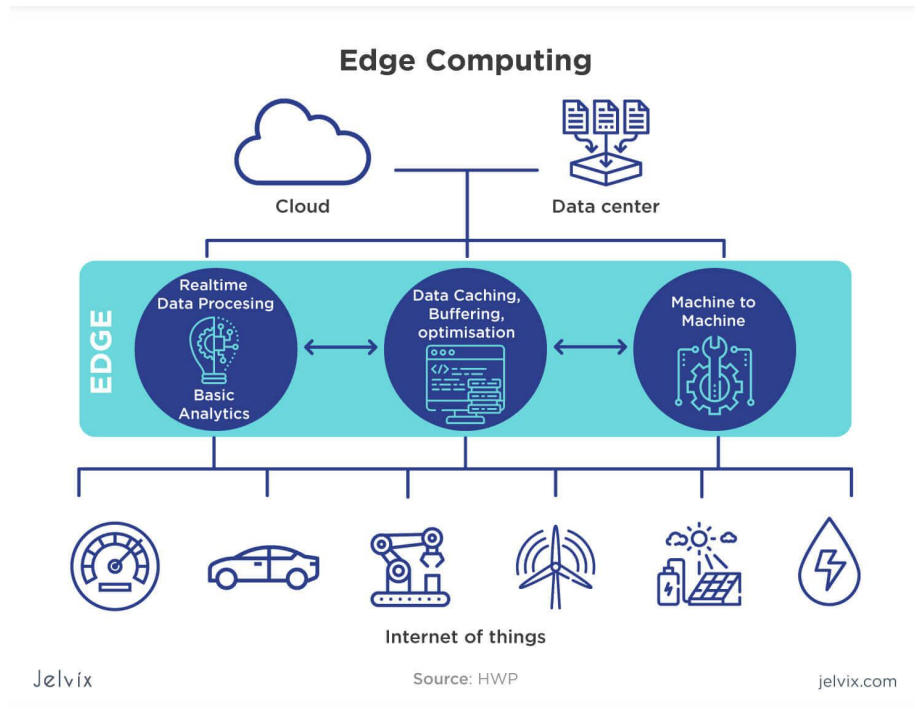
- **Edge Device:** Any device performing computation close to data source
- **Edge Server:** Processes data near the source
- **IoT Gateway:** Connects devices to Cloud. Has extra capabilities (data routing, protocol conversion, analytics)

Smart ICU Monitoring System

Components:

1. IoT Devices

- Patient monitors (for heart rate, oxygen saturation, BP), ventilators, infusion pumps.



Smart ICU Monitoring System

2. Edge Node (Edge Server/IoT Gateway) in ICU

- **Realtime Data Processing & Basic Analytics:**
 - Continuously processes vital signs.
 - Detects anomalies like a sudden drop in oxygen levels and triggers alerts immediately to nurses or doctors.
- **Data Caching, Buffering, Optimization:**
 - Stores recent data locally to avoid network overload.
 - Sends only summarized health logs or anomaly reports to the hospital's cloud system.
- **Machine-to-Machine (M2M) Communication:**
 - If a patient's oxygen drops suddenly, the monitor can signal the ventilator to increase oxygen delivery without waiting for cloud instructions.

Smart ICU Monitoring System

3. Cloud/Data Center

- Stores full patient history.
- Runs long-term analytics, risk modeling, and supports remote specialists for diagnosis.

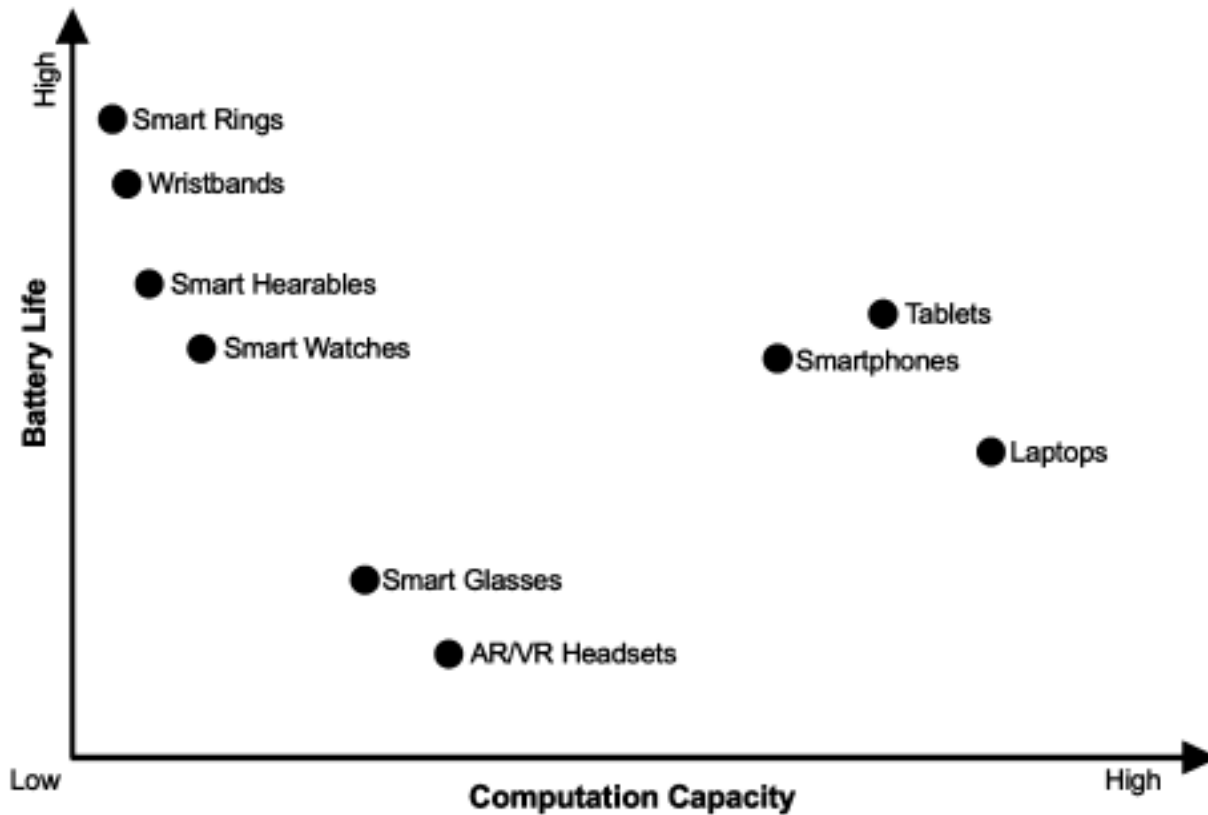
Edge Computing: Paper

Technical specifications of some COTS wearable devices.

Product name	Product type	Weight (g)	Battery	Connectivity		
				Cellular	Wi-Fi	Bluetooth
Apple Watch Series 6	Smart Watch	30.5–47.1	18 hours	✓	✓	✓
Fitbit Charge 4	Wristband	30.1	7 days			✓
Microsoft HoloLens 2	Mixed Reality Headset	566	2–3 hours		✓	✓
Bose Frames Tempo	Smart Glasses	371.4	8 hours			✓
Amazon Echo Loop	Smart Ring	9–12	1 day			✓
Mi Smart Band 5	Wristband	11.9	>14 days			✓
Apple AirPods Max	Smart Hearable	384.8	20 hours			✓
Huawei X Gentle Monster Eyewear II	Smart Glasses	44.2–48.1	5 hours			✓

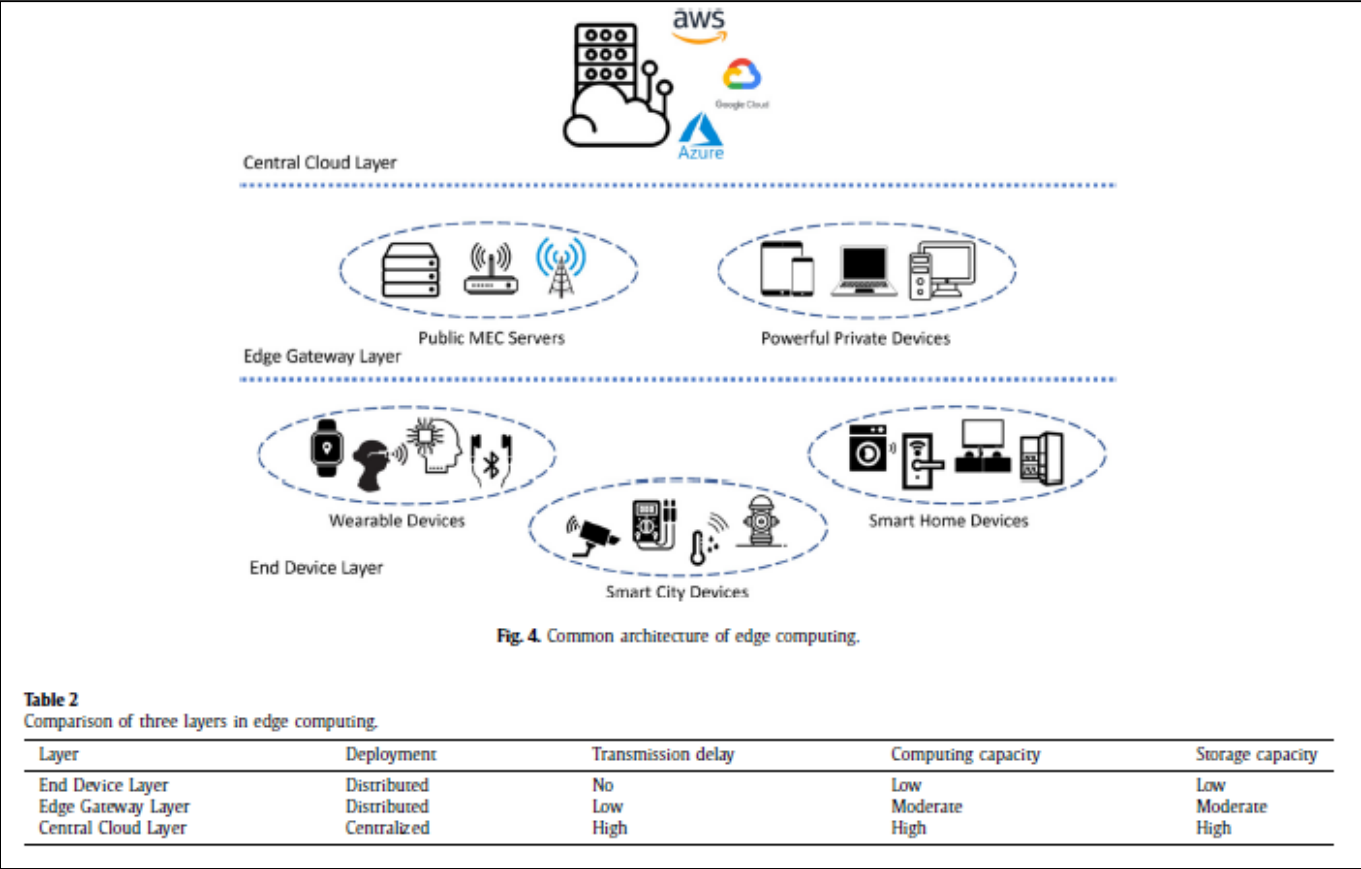
A survey on edge computing for wearable technology by Jin, Li et al., Digital Signal Processing, 2021
 URL - https://www.cse.msu.edu/~lilingk1/publications/DSP_Survey_Edge_Comp.pdf

Edge Computing: Paper



A survey on edge computing for wearable technology by Jin, Li et al., Digital Signal Processing, 2021
URL - https://www.cse.msu.edu/~lilingk1/publications/DSP_Survey_Edge_Comp.pdf

Edge Computing: Paper



MEC -> Multi-Access Edge Computing
A survey on edge computing for wearable technology by Jin, Li et al., Digital Signal Processing, 2021
URL - https://www.cse.msu.edu/~lilingk1/publications/DSP_Survey_Edge_Comp.pdf

Example I

- Autonomous vehicle edge computing devices collect data from cameras and sensors on the vehicle, process it, and make decisions in milliseconds, such as self-parking cars

Following devices collect and process data in real time on the vehicle

- NVIDIA Jetson AGX Orin / Xavier - AI-powered edge computing platform
- LiDAR Units (e.g., Velodyne LiDAR) - Detects objects and distance
- Onboard ECU (Electronic Control Unit) - Processes sensor input and executes vehicle control
- Radar Sensors - Detect speed and distance of nearby vehicles
- Camera Modules (e.g., Mobileye EyeQ) - Visual input for lane detection, signs, and objects

Example II

- In order to accurately assess a patient's condition and foresee treatments, data is processed from a variety of edge devices connected to sensors and monitors.

Following devices collect patient vitals and process them near the point of care

- Raspberry Pi with medical-grade sensors - Used in prototyping or remote health monitoring
- Intel NUC or NVIDIA Jetson Nano - For local AI-powered health analytics
- Smart Patient Monitors (e.g., Philips IntelliVue) - Monitors vitals like ECG, SpO₂
- Wearables (e.g., Fitbit, Apple Watch) - Tracks heart rate, activity, SpO₂

Key benefits of Edge for IoT

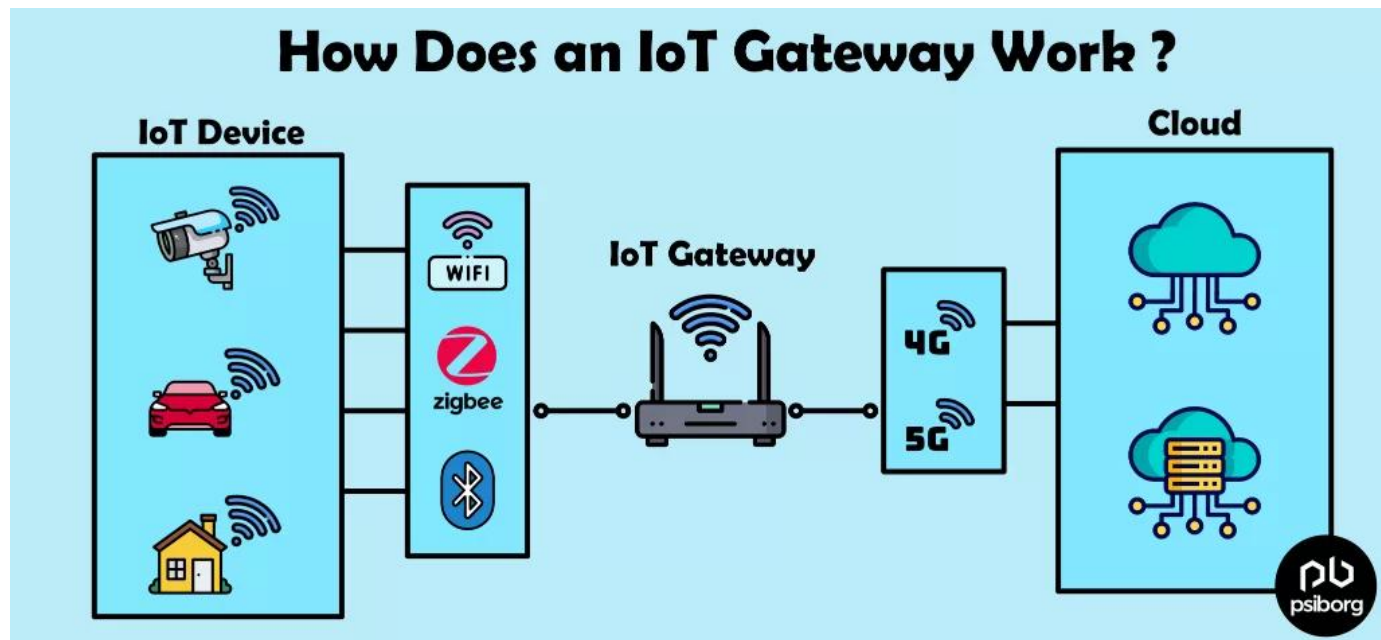
- Low latency (low delay in milliseconds)
- Longer battery life for IoT devices
- Access to data analytics and AI [Ex: Azure Public Multi-Access Edge Compute (MEC)]
- Resilience (reduced failures)
- Scalability
- More efficient data management



IoT Gateway

IoT Gateway

- An IoT gateway is a centralized hub that connects IoT devices and sensors to cloud-based computing and data processing.
- Modern IoT gateways often allow bidirectional data flow between the cloud and IoT devices.



IoT Gateway Key Functions

1. **Data Aggregation:** Collects data from various IoT devices and consolidates it for processing.
2. **Protocol Translation:** Bridges different communication protocols (e.g., MQTT, CoAP, HTTP, Bluetooth)
3. **Data Processing:** Performs preliminary data processing at the edge, reducing the volume of data sent to the cloud for further analysis and saving bandwidth.
4. **Security:** Acts as a secure layer, implementing encryption and security measures to protect data as it moves between devices and cloud services.
5. **Connectivity Management:** Maintains reliable communication between IoT devices and backend systems, ensuring continuous data flow despite potential network fluctuations.
6. **Edge Computing Capabilities:** Supports local processing and analytics for quick decision-making and reduced latency, which is particularly useful in time-sensitive applications.

IoT Gateway (Software)

Platform	Provider	Functionality
AWS IoT Greengrass	Amazon	Allows local device processing, AWS Lambda functions at the edge
Azure IoT Edge	Microsoft	Runs containers and ML models on edge devices, integrates with Azure
BalenaOS + balenaCloud	Balena	Lightweight OS and management platform for IoT devices
Kaa IoT Gateway	KaaloT	Open-source middleware with edge routing and transformation features
ThingsBoard Gateway	ThingsBoard	Collects and converts device data to communicate with cloud platforms



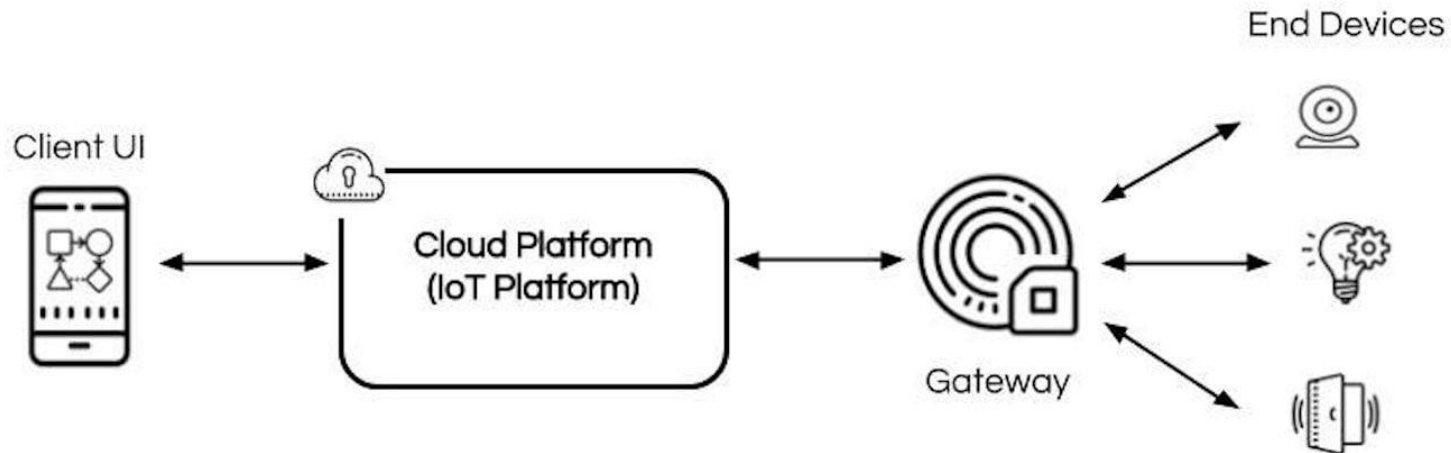
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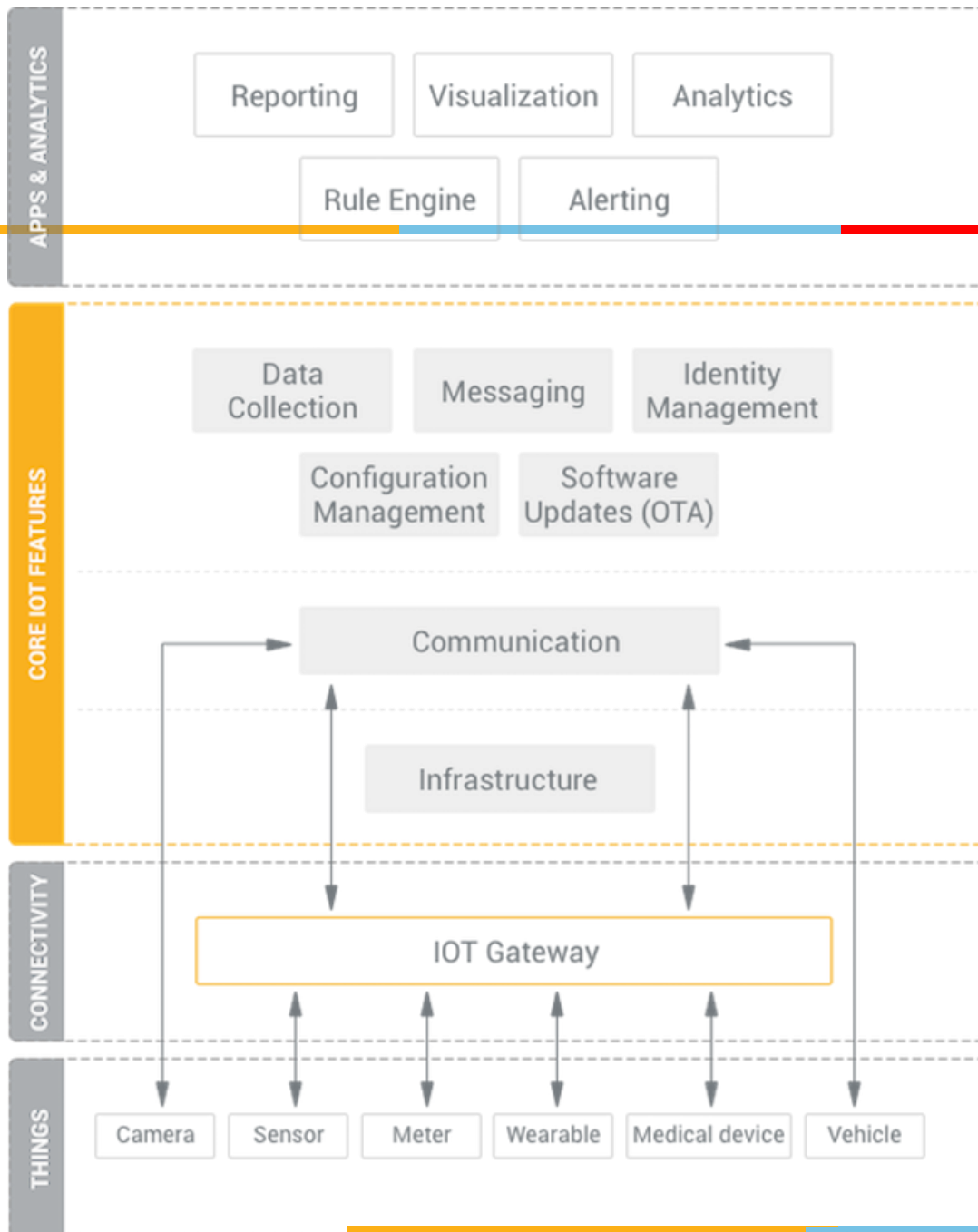
IoT Platforms

IoT Platform

- An IoT platform is a **multi-layer technology** that enables provisioning, management, and automation of connected devices within the Internet of Things universe.
- It acts as a “Middleware Technology”



Ref - <https://www.kaaiot.com/blog/what-is-iot-platform>



IoT Platform Technology Stack



IoT Platforms

1. OpenRemote [Open Source] -> Real Use Case in Practice
2. AWS IoT Architecture [Commercial] -> Understand the AWS Resources and Architecture
3. ThingsBoard [Open Source] -> Build simple use cases from Scratch

List of other popular IoT Platforms are provided in URL

<https://www.opensourceforu.com/2023/03/ten-popular-iot-platforms-you-should-be-aware-of/>

Case Study: Bipolar Disorder

- Bipolar Disorder (BD) is a recurrent chronic disorder characterized by fluctuations in mood state and energy, which affects over 1% of the World population.
- BD is a primary cause of disability among people, leading to functional and cognitive impairment, with increased morbidity, especially death by suicide.
- Compared to a normal, mentally-stable individual, an individual suffering from Bipolar Disorder experiences extreme mood fluctuations, classified into “**manic episodes**” and “**depressive episodes**”, which typically last between days to months.
- While the manic episodes are characterized by racing thoughts, feeling of elation, extreme irritability etc., the depressive episodes are characterized by feelings of extreme sadness, restlessness, trouble in concentration, insomnia etc.
- URL -> [https://en.wikipedia.org/wiki/3_\(2012_Indian_film\)](https://en.wikipedia.org/wiki/3_(2012_Indian_film))

Case Study: Bipolar Disorder

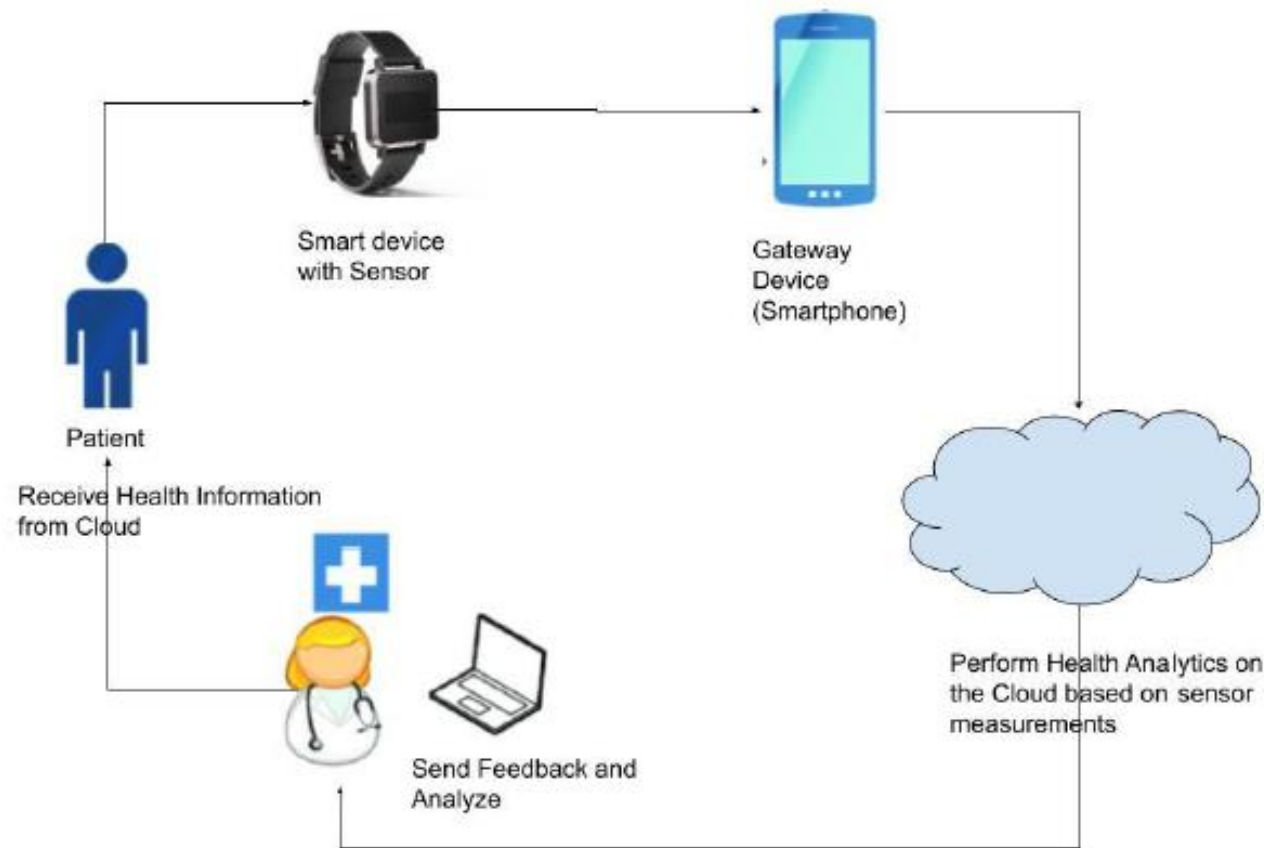


Fig 1. Smart Healthcare System for Monitoring Bipolar Disorder



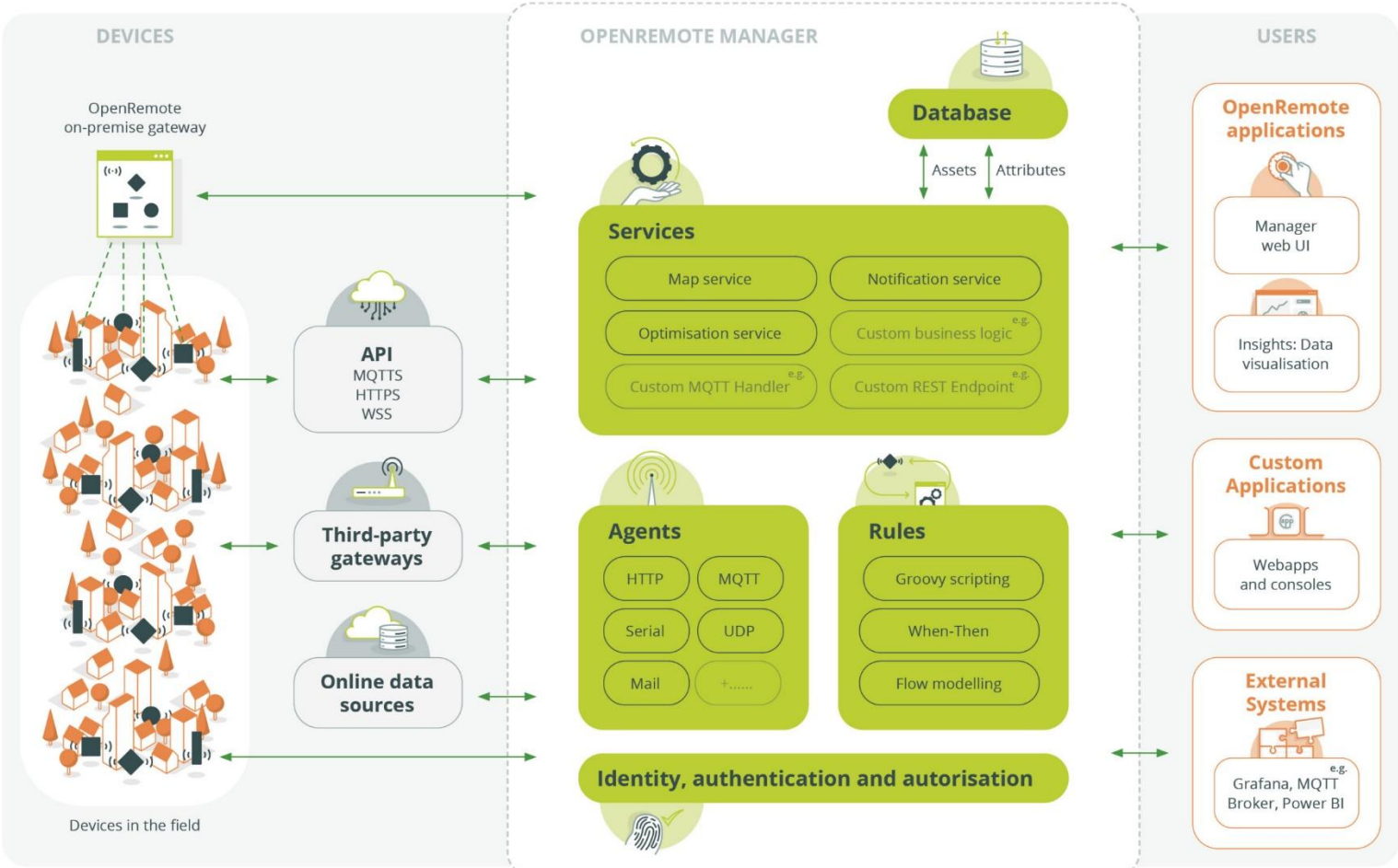
OpenRemote IoT Platform

OpenRemote

1. OpenRemote is an open-source platform for building IoT (Internet of Things) and smart management applications.
2. It provides a flexible and customizable solution for connecting and managing devices, sensors, and data in various types of smart environments.
3. These environments can include smart homes, cities, energy management systems, and industrial automation.
4. Supports communication protocols such as MQTT, HTTP etc.

* **MQTT (Message Queuing Telemetry Transport)** is a lightweight, publish-subscribe messaging protocol. Promotes topic-based messaging. Devices subscribe to specific topics and receive messages only related to those topics.

OpenRemote Architecture



OpenRemote – SmartCity Application

Smart City: Monitor your city assets and optimize public services

Predict traffic flows and define profiles

URL for OpenRemote Demo -> <https://www.openremote.io/demo/>

Username - smartcity

Password - smartcity

OpenRemote – SmartCity Application

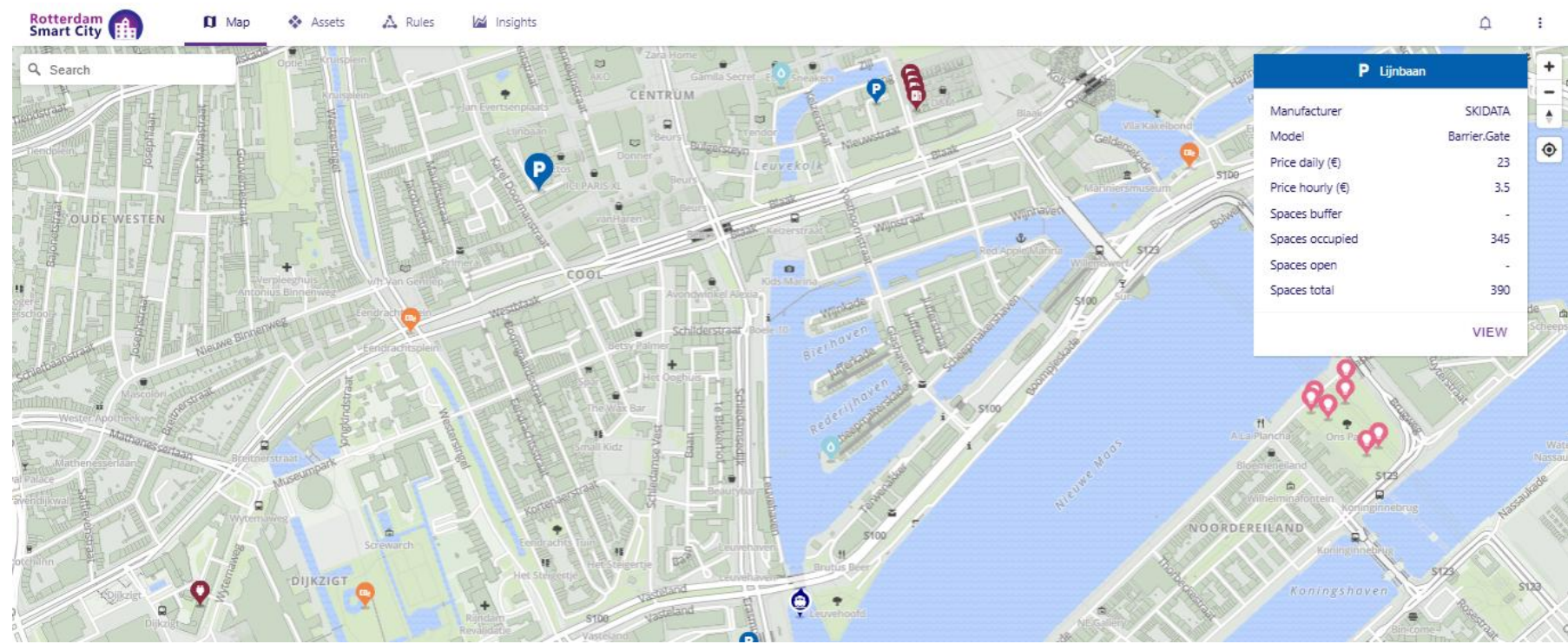
Scenario

We are considering / analyzing the “Parking Spaces” related data within SmartCity app.

1. **Map** -> Select “P” which stands for “Parking”. Click on “View” to see the Asset data
2. **Asset** -> View “Lijnbaan”, “Erasmusbrug” and “Markthal” parking asset’s attributes (values)

** These data in the “parking spaces”, can be captured through “Ground Sensors” which detects cars, overhead cameras which capture entire park spaces, Smart Parking Meters (payment details), or ANPR (Automatic Number Plate Recognition) Systems, installed at entry or exit points

Step 1 – Map



Step 2 - View Asset



Rotterdam Smart City

Map

Assets

Rules

Insights

Assets

Filter...

Consoles

Energy management

Environment monitor

Mobility and safety

Lighting Noordereiland

Parking group

Erasmusbrug

Lijnbaan

Markthal

Ship group

Stationsplein

Simulator agent

Lijnbaan

Created: Nov 15, 2024 8:30 AM

INFO

Manufacturer

SKIDATA

Updated: Nov 15, 2024 8:30 AM

Model

Barrier.Gate

Updated: Nov 15, 2024 8:30 AM

Notes

Updated: Nov 15, 2024 8:30 AM

ATTRIBUTES

Price daily (€)

23

Updated: Nov 15, 2024 8:30 AM

Price hourly (€)

3.5

Updated: Nov 15, 2024 8:30 AM

LOCATION

Updated: Nov 15, 2024 8:30 AM

HISTORY

Attribute

OpenRemote – SmartCity Application

Scenario

We are considering / analyzing the “Parking Spaces” related data within SmartCity app.

1. **Rules** -> “When-Then” scenario OR “Flow”
 - a. Select “Parking: Almost Full” and select “When-Then”
 - b. Select “Occupied Spaces” and select “Flow”

Step 3 – Rules -> “When-Then”

Rotterdam Smart City

Map

Assets

Rules

Insights

Rules

De Kuip

De Rotterdam: Battery use

De Rotterdam: Power balance

Environment monitoring: Alerts

Euromast

Light group: On/Off

Markthal

Markthal: All chargers in use

Ons Park: Brighten lights

Ons Park: Dim lights

Parking: Almost full

Parking: Occupied spaces

Station: Crowded square

Rule name*

Parking: Almost full

ID: 1012

Enabled ☒

ALWAYS ACTIVE

SAVE

When...

P

Asset

Lijnbaan

Attribute

Spaces occupied

Operator

Greater than

Spaces occupied

40

+ ADD ATTRIBUTE

+ ADD CONDITION

Or when...

+

Then...

ALWAYS

Recipients

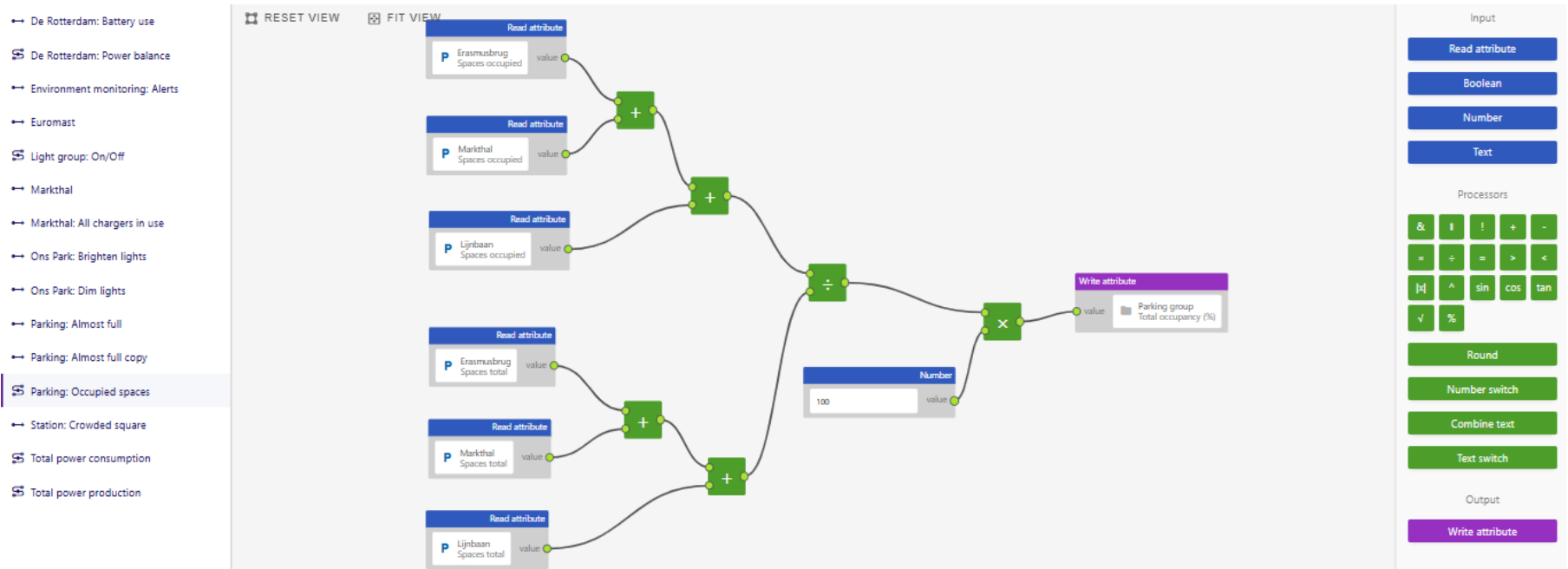
Custom attribute

shreyassureshrao@gmail.c

MESSAGE

+ ADD ACTION

Step 4 – Rules -> “Flow”



Calculate -> Parking Group Total Occupancy %

Formula -> Spaces Occupied (E+M+L) / Spaces Total (E+M+L)
 => 1246 / 1275 = **97.7%**

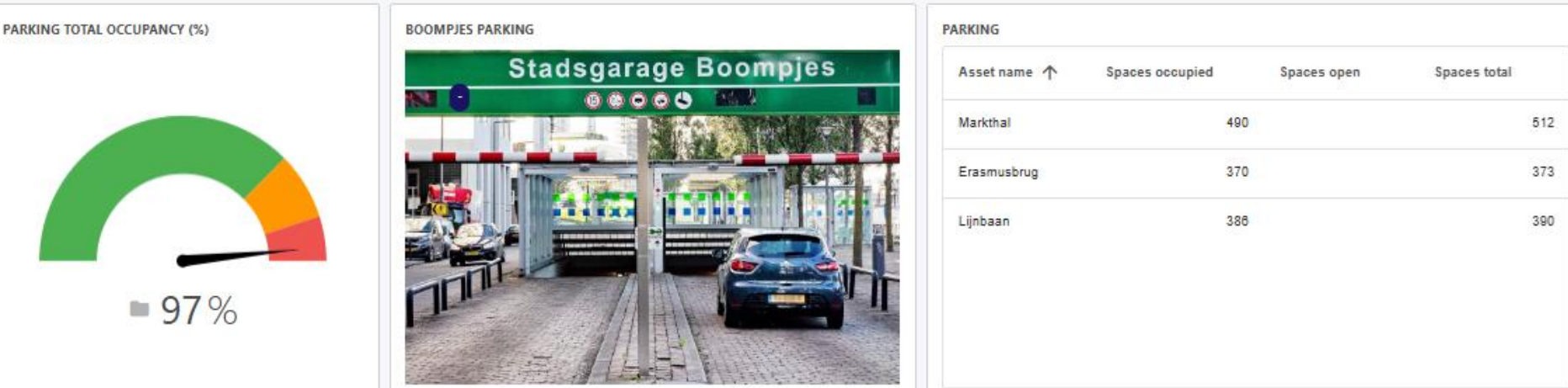
Step 5 – Insights

Select “Parking Dashboard”



Step 5 – Insights

Select “Parking Dashboard”



Calculate -> Parking Group Total Occupancy %

Formula -> Spaces Occupied (E+M+L) / Spaces Total (E+M+L)
=> 1246 / 1275 = **97.7%**



Thank You!